TUWAIQ ACADEMY · AI DIPLOMA

Course 3 — Scalable Data Science

Does El Niño Reach Saudi Arabia?

Quantifying ENSO Impact on Temperature and Rainfall using GPU-Accelerated Machine Learning

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Abstract

ENSO is the largest source of year-to-year climate variability on Earth, yet the Arabian Peninsula is usually left off its impact maps. This study tests whether that omission is justified, using 1,095,720 hourly records for five Saudi cities (2001–2025) drawn from ERA5, NASA POWER and NOAA's RONI index. The pipeline ran on GPU with RAPIDS cuDF and cuML across 315 training runs under fifteen walk-forward windows. The answer turns out to depend entirely on which variable is asked about. Temperature shows nothing: four algorithms all returned negative R^2 from ENSO indicators alone, matching a direct correlation of −0.029. Rainfall shows a clear effect: rain days are 1.68× more frequent during El Niño ($\chi^2 = 145.5$, $p < 10^{-33}$), the signal survives three months ahead of the rain, and a classifier restricted to advance-available inputs reaches AUC = 0.72. Annual rainfall rises 10–74 mm depending on city, peaking in autumn at 2.41×. A mechanism test tied the seasonal pattern to moisture transport ($r = 0.93$) and, decisively, the same mechanism runs backwards in summer. Three forms of target leakage were found and corrected during development; uncorrected, they would have reported $R^2 = 0.655$ instead of 0.200.

1. Introduction

Every two to seven years the equatorial Pacific changes temperature. Warming is called El Niño, cooling La Niña, and the coupled ocean–atmosphere phenomenon is ENSO. An ocean basin that size cannot shift temperature without reorganising global circulation, so the effects reach far beyond the Pacific — a mechanism known as teleconnection.

Those effects are well documented for Australia, Indonesia and Peru, and are forecast operationally. The Arabian Peninsula is usually absent from such maps, or marked as a region of weak and uncertain response. Whether that reflects a genuine absence or simply a gap in the literature is an open question.

It matters because the margins are thin. Saudi Arabia receives 33–111 mm of rain per year depending on location. A shift of a few tens of millimetres is not a rounding error at that scale — it is a substantial fraction of the annual total, and it bears on flood preparedness and water planning. NOAA publishes RONI monthly together with forward projections, so if the index carries usable information at a multi-month horizon, it becomes an input to seasonal outlooks rather than a retrospective label.

Two kinds of El Niño. Warming does not always occur in the same part of the Pacific. Canonical events warm the east; Modoki events warm the centre. Different warming locations produce different global wind patterns, so they affect different regions. Published work on the Arabian Peninsula attributes the rainfall response of recent decades specifically to Modoki events — a claim this study is positioned to test directly, since its data fall entirely within the period that literature describes.

Problem Statement

Given ENSO indicators available in advance, predict (a) the hourly temperature anomaly and (b) whether measurable rain falls on a given day, across five Saudi cities.

Each task runs twice — once with ENSO indicators alone, once with concurrent atmospheric variables added. The first configuration is genuinely predictive; the second is diagnostic. The gap between them is itself a result: it separates what can be anticipated from what requires knowing the present state of the atmosphere.

Objectives

- Build a GPU-accelerated pipeline processing over one million climate records end to end.
- Establish any ENSO effect statistically before modelling, so conclusions do not rest on model output.
- Compare four algorithms on identical data under time-series-appropriate validation.
- Determine whether a detected effect precedes the outcome, giving a usable forecast horizon.
- Measure GPU utilisation, memory and execution time against model complexity.
- Identify the physical mechanism behind any effect rather than reporting correlation alone.

2. Data and Preprocessing

Source	Variables	Notes
ERA5 via Open-Meteo	15	ECMWF reanalysis — observations assimilated into a physics model, giving gap-free hourly coverage
NASA POWER	3	MERRA-2 and CERES satellite radiation; hourly records begin 2001, which fixed the study period
NOAA CPC	RONI	Official ENSO index since Feb 2026; subtracts tropical-mean SST to separate ENSO from background warming

Copernicus CDS was the intended source. A single request took 42 minutes, projecting to roughly 63 hours for the full retrieval, so Open-Meteo was used instead — the same ERA5 data through a faster interface.

City	Latitude	Longitude	Setting
Riyadh	24.6877	46.7219	Central, inland
Jeddah	21.4858	39.1925	Western, Red Sea coast
Makkah	21.4225	39.8262	Western, inland, mountainous
Madinah	24.4686	39.6142	North-western, inland
Dammam	26.4207	50.0888	Eastern, Gulf coast

Five cities × 9,131 days × 24 hours = 1,095,720 records across 26 columns. The cities span coastal and interior settings on both the Red Sea and Gulf sides, which later proves important — the response is not uniform across the country. Data quality: zero missing values, zero duplicates on (city, time), all values within physical range.

2.1 The Anomaly Target

Climatology is the normal value for a place and time — the mean temperature in Riyadh in July at 2 pm across twenty years, for instance. The anomaly is the deviation from it. If Riyadh records 43.8 °C where 41.3 is normal, the anomaly is +2.5.

Predicting temperature directly would be worthless here. A model doing so would learn that summer is hot and afternoons are hot, score extremely well, and reveal nothing about ENSO. Subtracting climatology strips out what is already known and leaves the unexplained variance — the only place an ENSO effect could live.

Climatology is grouped by (city, month, hour), removing three known cycles: geographic, seasonal, and diurnal. The hour term is not optional. The day–night swing is 12–14 °C, larger than any plausible ENSO effect, and leaving it in the target caused one of the three leaks documented in Section 3.2. Climatology is computed from training years only, inside each fold.

2.2 Feature Engineering

RONI lags at 1, 3 and 6 months were added, since teleconnections propagate over months. Lags were built on a month-level table rather than the hourly frame, because shift() operates by row position — applied directly it would have shifted by hour. Dropping the first six months of 2001 for want of a six-month lag left 1,074,000 rows. City was one-hot encoded into five binary columns.

Rainfall needed a different treatment entirely. At hourly resolution 98.67% of precipitation values are zero — the 25th, 50th and 75th percentiles are all zero. Regression explains variance, and a near-constant target has none, so any regression result would reflect the target's structure rather than anything about ENSO. Data were aggregated to daily totals (44,750 rows) with a binary target at 0.1 mm, the standard threshold for measurable rain. The positive class rises from 1.3% of hours to 7.2% of days.

2.3 Validation

Random splitting is invalid for time series — it can train on 2022 and test on 2015. Walk-forward validation with an expanding window was used: train on all years before Y, test on Y, advance. Fifteen windows spanning 2011–2025, beginning at 2011 so the narrowest window retains ten years of training.

3. Methodology

3.1 How the Study Was Built

Two phases. An exploratory notebook established the methodology — testing feature combinations, running the ablation study, and uncovering the three leaks below. Once the correct approach was settled, the analysis was rebuilt from scratch rather than patched. This was deliberate, not tidying: the intermediate files from exploration carried two of the three leaks, so reusing them would have reintroduced exactly what the rebuild was meant to remove.

Statistics before models. For both targets, the relationship was tested statistically before any model was trained — phase comparison and chi-square for rainfall, direct correlation for temperature. Models were then run to confirm, not to discover. This ordering removes the strongest objection available to a reader: that a model may have learned an artefact. When the rainfall classifier later reached AUC = 0.62, it was confirming a difference already established at $p < 10^{-33}$.

3.2 Three Leaks

Each of the following would have inflated reported performance. Each was caught by feature importance, not by reading the code.

#	Leak	Mechanism	R^2 before fix
1	Global climatology	Baseline computed over all years, so the target itself carried test-period information	0.655
2	Derived variables	ET0 is computed from temperature via Penman–Monteith; it alone drove 67% of decisions	0.580
3	Diurnal cycle	Climatology omitted hour, leaving the day–night swing in the target; solar radiation then drove 60%	0.175

> The reported figure fell from 0.655 to 0.175. But the contribution of RONI features to model decisions rose from 0.66% to roughly 20% over the same three corrections. The signal under study only became visible once the noise around it was removed.

3.3 Models and Ablation

Model	Library	Task	Role
Linear / Logistic Regression	cuML	Both	Baseline
Random Forest	cuML	Both	Non-linearity and interactions
KNN Regressor	cuML	Regression	Instance-based; different inductive bias
XGBoost	xgboost	Both	Sequential boosting; outside cuML

Algorithm diversity was intentional. Methods with different inductive biases converging on the same answer make that answer a property of the data rather than of one algorithm. Hardware: NVIDIA Tesla T4, 15,360 MB. Total: 315 training runs.

Two feature sets define the two questions. ENSO_only (9 features) contains everything known in advance, since NOAA publishes RONI forecasts. Full (21 features) adds concurrent atmospheric variables — forecasting next month with it would require knowing next month's pressure. For rainfall a third set, ENSO_season (11 features), adds calendar month as cyclical terms; month is known in advance, so the set stays predictive. Four temperature-derived variables were excluded from Full for the reason given above.

Ablation step	Mean R^2	Decision
Corrected baseline	0.175	—
+ City as feature	0.191	Retained
+ Additional Niño indices (1+2, 3, 4, 3.4)	0.187	Discarded
+ Random Forest tuning (200 trees, depth 16)	**0.202**	Retained
+ RONI_lag9	0.198	Discarded

The extra Niño indices added nothing, and this has an explanation rather than being noise: RONI is itself derived from the Niño 3.4 region, so those indices largely duplicate information already present.

4. Results

4.1 Metrics

R^2 measures variance explained; negative means worse than always predicting the mean. RMSE is error in °C against a target standard deviation of 2.37. AUC is the probability a rainy day scores above a dry day, with 0.5 meaning no skill.

Accuracy is reported for rainfall but not used, and the reason is worth stating. With 7.2% of days wet, predicting "no rain" always scores 92.5%. In the results below, all six accuracy values across three models and two feature sets are identical at 0.9247 — exactly the proportion of dry days. A metric that cannot separate AUC 0.575 from 0.724 is measuring nothing.

4.2 Temperature: Nothing There

Model	R^2 ENSO_only	R^2 Full	RMSE Full
Random Forest	−0.133	**0.200**	**2.108**
XGBoost	−0.133	0.176	2.128
KNN	−0.769	0.160	2.161
Linear Regression	−0.058	0.117	2.214

Every model is negative with ENSO_only. The best result overall — 0.200 — comes entirely from concurrent variables that cannot be known ahead of time.

The pattern of failure is the finding. Ranked on ENSO_only: Linear (−0.06) > RF ≈ XGB (−0.13) ≫ KNN (−0.65). Performance degrades monotonically with model capacity. Linear regression is too constrained to do much harm. A 200-tree forest at depth 16 memorises training-year noise. KNN, which does not generalise at all, collapses — and for a structural reason: RONI is monthly, so roughly 720 hourly rows per city-month share identical feature values, and the ten nearest neighbours all come from the same month with unrelated targets. Give a model more freedom in the absence of signal and it fits more noise. That is what this ordering means.

None of this was a surprise. The direct correlation between RONI and monthly temperature anomaly is −0.029, with lags at −0.023, −0.002 and −0.002. Per city the largest magnitude is Makkah at −0.075 — under 0.6% of variance — and the sign flips between western and eastern cities. That test involves no model, no split and no hyperparameters, so the negative R^2 confirms an earlier finding rather than introducing a new one.

Aggregating to monthly means (1,470 rows) was tested as a robustness check, in case hourly anomalies were the wrong scale for a phenomenon that develops over months. R^2 becomes more negative still, but those values are not comparable — the sample fell by three orders of magnitude and Full collapsed alongside ENSO_only, which cannot indicate absent signal given that the atmospheric variables demonstrably work at hourly scale.

4.3 Rainfall: A Real Effect

Phase	Rain days	Sample
El Niño	10.40%	8,690 days
Neutral	6.75%	18,905 days
La Niña	6.18%	17,155 days

Rain days are 1.68× more frequent during El Niño, with Neutral falling between the two — a gradient rather than a jump. Chi-square test of independence: $\chi^2 = 145.48$, $p = 1.68 \times 10^{-33}$.

City	El Niño	La Niña	Ratio	p-value	Significant
Jeddah	11.51%	6.06%	1.90×	1.0e−11	Yes
Makkah	18.01%	9.97%	1.81×	3.2e−16	Yes
Dammam	9.61%	5.65%	1.70×	1.9e−07	Yes
Riyadh	7.54%	5.10%	1.48×	5.7e−04	Yes
Madinah	5.35%	4.14%	1.29×	0.057	No

Madinah misses the 0.05 threshold and is not treated as confirmed. It is the driest city with the fewest rain days, so the test lacks power there — reported as inconclusive rather than quietly folded into the aggregate.

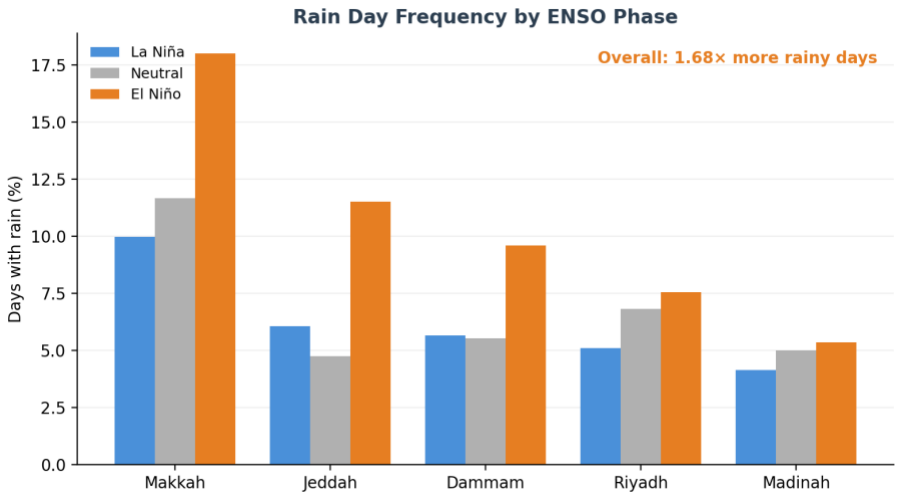


Figure 1. Rain day frequency by city and ENSO phase.

4.4 Is It Really ENSO, or Just the Season?

Saudi rainfall is strongly seasonal. If El Niño events happened to land in wet winters, the aggregate result above would appear without ENSO doing anything. Testing within each season separately settles it.

Season	El Niño	La Niña	Ratio
Autumn (SON)	11.27%	4.68%	**2.41×**
Winter (DJF)	14.27%	9.35%	1.53×
Spring (MAM)	10.56%	7.22%	1.46×
Summer (JJA)	1.23%	2.32%	0.53×

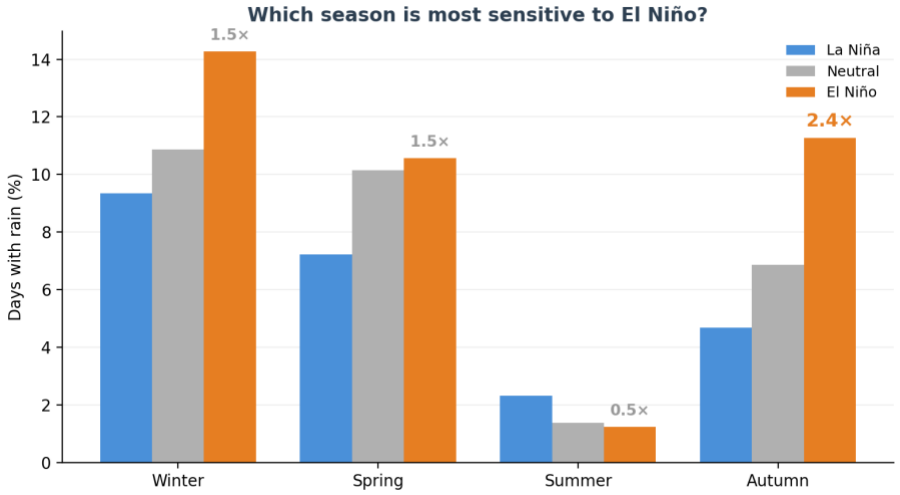


Figure 2. Winter has the highest absolute rain frequency; autumn has the strongest relative response.

The effect survives in every season but summer, and the autumn ratio holds across all five cities individually (2.11 to 2.61). Magnitude and sensitivity point to different seasons: winter has the most rain days in absolute terms but a lower ratio, because winter is wet regardless of ENSO. Autumn is dry by default, so El Niño more than doubles it. Summer reverses, on numbers too small to carry weight alone — but Section 4.7 shows the reversal is not noise.

4.5 How Much, and How Far Ahead

City	La Niña (mm)	El Niño (mm)	Increase	%
Dammam	61.2	113.5	+52.2	**+85.3**
Makkah	100.9	175.0	**+74.1**	+73.5
Jeddah	54.9	94.4	+39.5	+72.0
Riyadh	46.9	62.8	+15.9	+34.0
Madinah	37.2	47.6	+10.4	+27.8

Ranking depends on the measure — Dammam leads proportionally, Makkah in absolute terms — so both are given. Percentages inflate small baselines, which matters more in autumn, where Madinah's +152% amounts to 4.9 mm while Makkah's smaller +124% is 22.6 mm. Year-level aggregation was examined but not used for inference: only two years qualify as El Niño by mean annual RONI, since events span two calendar years.

Horizon	El Niño	La Niña	Ratio	p-value
Concurrent	10.40%	6.18%	1.68×	1.7e−33
1 month ahead	11.28%	6.64%	1.70×	2.2e−37
3 months ahead	11.32%	6.94%	**1.63×**	1.3e−32
6 months ahead	7.74%	6.08%	1.27×	4.5e−07

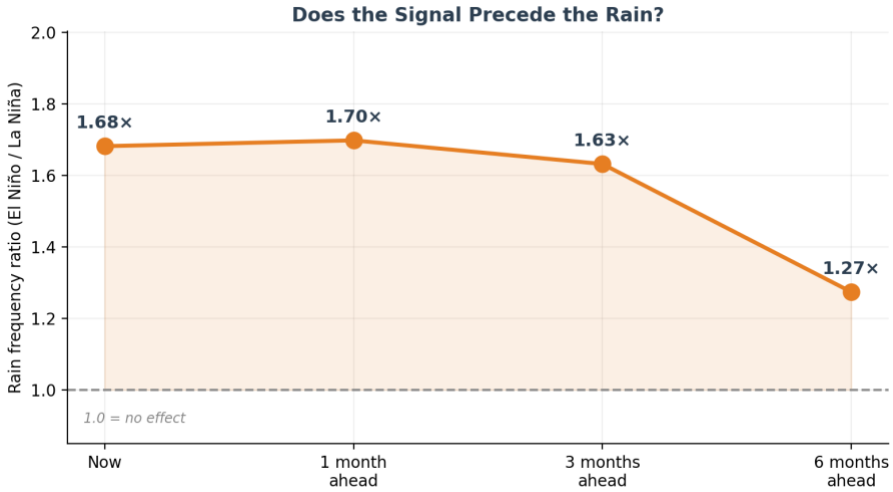


Figure 3. The signal holds almost fully at three months and remains significant at six.

This is what converts a description into a forecast. Western cities peak at lag 0–1 while eastern and central cities peak at lag 3 — Dammam reaches 1.95× three months ahead, higher than its concurrent value, suggesting the effect arrives inland with a delay.

One caveat stated plainly: *ENSO events persist for months, so RONI three months ago correlates with RONI now. Part of the lagged discrimination is persistence rather than delayed physical response. Operationally this changes little — the index is available in advance and separates wet periods from dry ones — but it should not be presented as a demonstrated lag mechanism.*

4.6 The Model

Model (AUC)	ENSO_only	ENSO_season	Full
Random Forest	**0.619**	**0.724**	**0.949**
XGBoost	0.593	0.695	0.948
Logistic Regression	0.575	0.695	0.916

Two numbers carry the conclusion. 0.619 is forecasting from ENSO alone. 0.724 adds calendar month, also known in advance — the operational figure. 0.949 is a diagnostic ceiling only: it includes same-day cloud cover, which is high precisely because rain is about to fall.

A reversal worth noticing. On temperature, Random Forest did worse than linear regression on ENSO_only. Here it does better. The same algorithms, the same procedure, opposite orderings — the difference being whether a signal exists at all. High capacity fits noise when there is nothing to learn and captures structure when there is. This is the clearest single piece of evidence that the temperature result was a property of the phenomenon, not a failure of method.

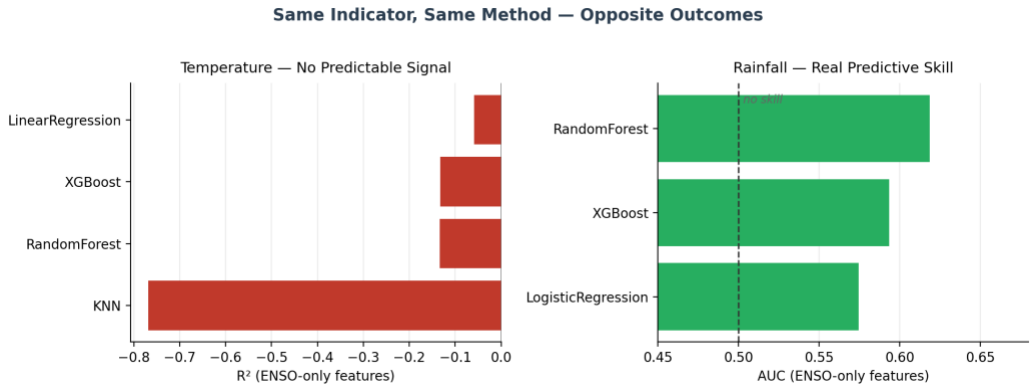


Figure 4. Identical inputs and validation: negative R^2 for temperature, above-chance AUC for rainfall.

SHAP. Contribution to the rainfall classifier splits as seasonality 39.4%, ENSO indicators 34.6%, city 26.0%. Within the ENSO group the concurrent index ranks highest (11.8%), followed by the three-month lag (10.2%) — consistent with the lead-time result. Direction matters as much as magnitude: mean SHAP is +0.064 during El Niño and −0.437 during La Niña, confirming the model learned a relationship with the physically correct sign rather than an arbitrary numerical association.

4.7 Why Autumn? A Mechanism Test

Autumn responds at 2.41× while winter reaches only 1.53×. Rather than leaving that as speculation, it was tested. The hypothesis: winter rain comes from Mediterranean systems that operate independently of ENSO, while autumn rain depends on moisture carried from remote sources — the pathway ENSO can modulate.

Season	Water vapour shift	Southerly wind shift	Rain ratio
Autumn	**+0.315**	**+0.156**	2.41×
Spring	+0.204	+0.060	1.46×
Winter	+0.041	−0.007	1.53×
Summer	**−0.321**	**−0.155**	0.53×

Shifts are in units of seasonal standard deviation, so the variables are comparable. Across the four seasons the correlation between moisture shift and rainfall response is r = 0.93. Winter shows essentially no wind response at all (−0.007), which is what the hypothesis predicts.

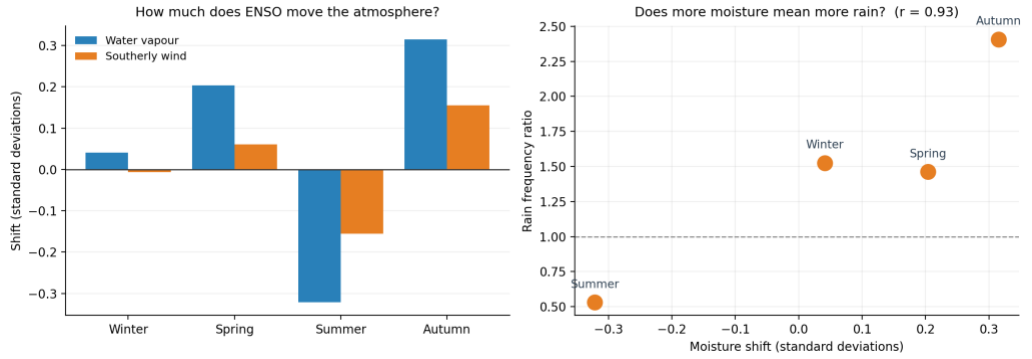


Figure 5. Left: how far ENSO displaces moisture and southerly flow each season. Right: that displacement against the rainfall response.

> Summer settles it. El Niño reduces water vapour (−0.321), reduces southerly flow (−0.155), reduces cloud — and rainfall falls to 0.53×. The same mechanism running backwards. An explanation that predicts the exception is worth more than one that only fits the rule.

Per city, all five show positive moisture and southerly wind shifts in autumn within narrow ranges, while winter signs are mixed. Riyadh has the largest autumn wind shift at +2.12 m/s — the most inland city, where moisture must travel furthest. The southerly direction points to sources south of the peninsula, distinct from the westerly Mediterranean pathway that brings winter rain.

Caveat: *the correlation rests on four seasonal points. It is indicative, not a statistical test, and moisture origin would need back-trajectory analysis on gridded fields to establish properly.*

4.8 Canonical or Modoki?

The literature attributes the recent Arabian rainfall response specifically to Modoki events. Our window falls entirely inside the period that claim describes, so it is directly testable. RONI is built on the Niño 3.4 region and does not distinguish the two types, so events were reclassified using the Niño 3, Niño 4 and Niño 1+2 indices. Two independent methods agreed on 98.3% of days.

City	Canonical	Modoki
Jeddah	**2.16×**	1.63×
Makkah	**2.03×**	1.58×
Dammam	1.38×	**2.03×**
Riyadh	1.31×	**1.66×**
Madinah	1.23×	1.36×
National	**1.71×**	**1.65×**

The published attribution is only partly supported. Nationally there is no distinction — both types produce a significant positive response ($p < 10^{-20}$). But the city-level split is orderly rather than random: western cities respond more to canonical events, eastern and central cities to Modoki. Two pathways whose signatures cancel when averaged, which also explains why eastern cities peaked at a longer lag in Section 4.5.

Caveat: *this approximates the El Niño Modoki Index using standard Niño boundaries, whereas published studies use EOF decomposition. The window also contains few events.*

5. GPU Performance

Comparison	CPU	GPU	Speedup
read_csv (1.07M rows)	5.169 s	0.229 s	**22.6×**
groupby (climatology)	0.095 s	0.008 s	11.4×
merge	0.223 s	0.025 s	9.1×
sort	0.367 s	0.023 s	15.9×
Random Forest training (50 trees)	**198.2 s**	**7.8 s**	**25.4×**
Linear Regression training	0.96 s	1.30 s	**0.74×**

The 25× on Random Forest is what made this study feasible — 315 training runs would have taken hours rather than minutes on CPU, and walk-forward validation would have been impractical. But linear regression is slower on GPU: its closed-form solution finishes faster than the cost of moving data to the device. Acceleration is justified by computational intensity, not by dataset size.

Training rows	CPU	GPU	Speedup
50,000	2.97 s	0.29 s	10.2×
200,000	13.34 s	0.64 s	21.0×
500,000	36.84 s	1.54 s	23.9×
810,960	73.59 s	2.73 s	**26.9×**

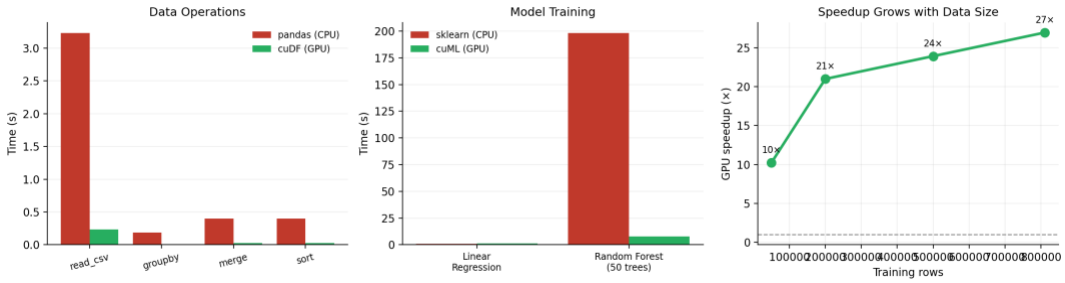


Figure 6. GPU versus CPU across data operations, model training, and increasing volume.

CPU time scales roughly linearly while GPU time grows slowly, so the advantage widens with volume — the specific justification for acceleration at this scale rather than in general.

Model	Mean util	Peak	Memory	Fit time
Random Forest	**97.1%**	100%	302 MB	54.4 s
Linear Regression	81.9%	100%	182 MB	1.2 s
XGBoost	78.3%	93%	264 MB	1.9 s
KNN	0.0%	0%	138 MB	0.1 s

Utilisation was sampled at 150 ms intervals by a background thread. Random Forest saturates the device. KNN reads zero because it performs no training — only storage — and finishes before a second sample can be taken. Memory stays under 2% of the 15 GB available, so compute rather than memory is the binding constraint.

Trees / Depth	Fit time	Predict time	Mean util
25 / 6	3.15 s	0.03 s	93.0%
100 / 10	16.55 s	0.05 s	97.6%
200 / 12	37.46 s	0.61 s	97.6%
200 / 16	52.85 s	**3.21 s**	97.4%
400 / 16	106.27 s	**7.93 s**	97.3%

Two different scaling laws in one table. Training time scales linearly with tree count — 200 to 400 doubles it. Inference time scales super-linearly with depth: raising depth from 12 to 16 at constant tree count increases prediction time more than fivefold, because leaf count grows exponentially. Depth is therefore far more expensive at inference than at training, which matters for deployment where inference cost recurs and training cost is paid once. Utilisation stays near 97% throughout, so these differences reflect real work rather than waiting.

6. Challenges

Data acquisition consumed the most time. Copernicus CDS rejected requests repeatedly with “cost limits exceeded” as the per-request limit was probed downward, and NetCDF proved more restrictive than GRIB. A successful request took 42 minutes. Parallel submission then revealed that client.retrieve() blocks until completion, so the apparent parallelism was sequential; a threaded rewrite was rejected with “queued requests temporarily limited”. Switching sources entirely was the answer, and produced a materially better pipeline than the original plan.

The leaks were the hardest part to accept. Three discoveries, each found through feature importance, each requiring a rebuild, cumulatively dropping R^2 from 0.655 to 0.175. It read as failure at the time. It was the point at which the work became defensible. The diagnostic signature was consistent throughout: when one feature dominates a model, the target is being recovered rather than predicted.

Metric selection. Initial rainfall results looked excellent at 93% accuracy until inspection showed every model predicting “no rain” universally. Identical accuracy across three different algorithms made it unmistakable.

Session volatility. Colab sessions reset repeatedly during long downloads. One reset produced a dataset containing only Riyadh, which propagated silently through several processing steps before being caught. A shape-and-city check was added after every stage in response.

7. Discussion

The answer depends on which variable is asked about, and that conditionality is the finding rather than a hedge. Temperature shows no usable relationship, established by four algorithms across fifteen windows, at two time scales, and by a correlation test requiring no model at all. Rainfall shows a clear one: 1.68× more rain days during El Niño, holding three months ahead, with AUC = 0.72 from advance-available inputs.

The two are physically coherent. Desert surface temperature is governed by local radiation balance; precipitation requires moisture brought from elsewhere. ENSO acts on the second pathway, and the mechanism test confirms it at r = 0.93 — including the summer reversal, where reduced moisture produces reduced rain.

What This Does Not Show

- Not a weather forecasting system. The models shift probabilities over months; they say nothing about specific dates.
- The 0.95 AUC is not forecast skill. It depends on same-day cloud cover, which is a consequence of imminent rain rather than a predictor of it.
- Temperature results do not make ENSO climatically irrelevant to Saudi Arabia — only that it does not modulate temperature anomalies at these scales.
- Annual-equivalent millimetres are illustrative. Real ENSO events last months, not years.
- Percentages need their absolute counterparts. Madinah's +152% autumn increase is 4.9 mm.

Limitations

- Five grid cells represent five cities. Regional patterns between them are invisible, and moisture origin cannot be traced.
- Fifteen variables come from ERA5 and three from NASA POWER (MERRA-2) — independent systems with different assimilation schemes. The NASA variables were excluded from final feature sets, which limits but does not eliminate the concern.
- ALLSKY_SFC_LW_DWN is downward longwave radiation where ERA5 provides net flux; used as an approximation.
- Twenty-five years contain few ENSO events, which constrains the event-type analysis in particular.
- Part of the lagged discrimination reflects ENSO persistence rather than a delayed physical response.

If This Were Deployed

The right output is a seasonal probability shift, not a categorical prediction. Autumn is the highest-value window and coastal regions the most responsive. Error costs are asymmetric — a false alarm means unnecessary preparation, a missed event in an area with limited drainage capacity means considerably worse — so thresholds should be set against that asymmetry rather than left at 0.5.

Future Work

- NAO and IOD as alternative indices — both are more frequently linked to Middle East precipitation than ENSO, and would test whether ENSO is the dominant remote driver or merely one of several.
- Gridded spatial analysis replacing point extraction, enabling impact maps and revealing gradients between cities.

- EOF-based canonical/Modoki classification, to test the spatial split found here with the method the literature uses.
- Extreme event targets — heavy rainfall and heat waves, more actionable than mean anomalies.
- Back-trajectory analysis on gridded fields to test the moisture pathway directly rather than by correlation.
- Operational prototype consuming live NOAA RONI forecasts to produce city-level seasonal outlooks.

8. References

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[11] [Arabian Peninsula winter rainfall and ENSO teleconnection study — complete this citation from the original source before submission.]

Note on [11]: *the study on long-term changes in Arabian Peninsula winter rainfall and its relationship to canonical and Modoki events, discussed in Sections 1 and 4.8. Full bibliographic details must be retrieved from the original publication.*